Narrow-Band Exclusion and the (D, ε) Phase Diagram

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Abstract

We combine the repetition-rigidity theorem of the Finite Spectral Shadows program with a ballot-type count of discrepancy-confined valuation words. The count grows at rate $\lambda(D)$, computed by strip transfer matrices, and crosses the band capacity 2 at $D^* = 1.246$ (multiplicative band ratio $2^{2D^*} \approx 5.6$). Consequence: a *divergent* positive Syracuse orbit cannot dwell in any band of ratio < 5.5 at height Y for more than $O(Y^{\theta(D)})$ steps, $\theta(D) = \ln \lambda(D)/\ln 2 < 1$ — unconditionally. For bands of ratio $\lesssim 2.1$ the same squeeze excludes cycles as well, via the Baker bound. Tilting the transfer matrix by the non-coboundary shadow observable extends λ to $\lambda_{\text{eff}}(D, \varepsilon)$ and produces a phase diagram in the (band-width, bias) plane whose excluded region covers all narrow bands and all wide bands of bias $\gtrsim 0.5$: counterexample windows are confined to wide bands with low local bias, so the global bias demanded by the keystone arrow must concentrate in band crossings.

1 The strip count

For a valuation word $w = (b_1, \ldots, b_p)$ write $S_j = \sum_{i \le j}(\log_2 3 - b_i)$ and call w *D-confined* if $|S_j| \le D$ for all $j \le p$. Let $N_D(p)$ denote the number of D-confined words of length p.

Lemma 1 (Strip growth rate)**.** *$N_D(p) \le C_D\, \lambda_\delta(D + \delta)^p$ for every grid step $\delta > 0$, where $\lambda_\delta(D)$ is the Perron root of the δ-coarsened strip transfer graph (states: grid points of $[-D, D]$; one edge per letter b with shift $\log_2 3 - b$, endpoints rounded outward). The bracketing is monotone in D and converges as $\delta \downarrow 0$. Computed values ($\delta = 10^{-2}$):*

D	0.5	0.8	1.0	1.232	1.5	2.0	3.0	∞
$\lambda(D)$	1.00	1.50	1.75	**2.00**	2.14	2.35	2.56	2.78

with $\lambda(\infty) = e^{h_0}$, h_0 the max entropy of letter laws of mean $\log_2 3$. We write $\theta(D) = \ln \lambda(D)/\ln 2$, so $\theta(D^*) = 1$ at $D^* = 1.232$.

Proof. Any D-confined path induces a path on the coarsened graph with widened strip; the number of length-p paths on a finite graph is at most $C\rho^p$ with ρ its Perron root. Monotonicity in D and in δ is immediate from edge inclusion. □

2 Narrow-band dichotomy

Recall the repetition-rigidity theorem: two positions of a positive orbit carrying the same length-p valuation block, lying within 2^{p+1} of each other, are equal (the block class is a single residue class modulo 2^{B_u+1}, $B_u \ge p$).

Theorem 1 (Narrow-band dichotomy). *Let an orbit segment of length M satisfy $x_n \in [Y, 2^{2D}Y]$ throughout, and set $p = \lceil \log_2(Y(2^{2D}-1)) \rceil$. If*

$$M \ > \ C_D\, \lambda(D)^p + p \ \asymp \ C'_D\, Y^{\theta(D)},$$

then the segment contains two equal points: the orbit is purely periodic — an integer cycle inside the band, of period $\le M$ and minimum $\ge Y$. Consequently:

(a) (Unconditional, divergent case.) *A divergent orbit is acyclic, so it can spend at most $C'_D Y^{\theta(D)}$ consecutive steps in any band of ratio $< 2^{2D^*} \approx 5.6$ at height Y.*

(b) (Full exclusion at very narrow bands.) *If $\theta(D) < 1/(\tau+1)$ (with $\tau = 13.3$ from the Baker cycle bound $x_{\min} \le Cp^{\tau+1}$; numerically $D \lesssim 0.55$, band ratio $\lesssim 2.1$), the manufactured cycle has $Y \le C(C'_D Y^{\theta})^{\tau+1} < Y$, a contradiction:* no *orbit, periodic or not, dwells beyond the bound.*

(c) (Intermediate regime.) *For $\theta \in [1/(\tau+1), 1)$ the residual object is a high cycle with period pinned to $[(Y/C)^{1/(\tau+1)}, C'_D Y^{\theta}]$ — the classical cycle problem, quantitatively localized.*

Proof. The segment's word is D'-confined with $D' \le D + o(1)$ (the $\log_2(1+\frac{1}{3x})$ corrections are $O(M/Y)$). Its length-p blocks are $2D$-confined words, of which there are at most $C\lambda(2D \to D\text{-bracketing})^p$; with the stated M the pigeonhole yields two positions with equal blocks, and the band diameter $Y(2^{2D}-1) < 2^{p+1}$ forces equality by repetition rigidity. The corollaries follow as stated. □

Corollary 1 (Bounded-discrepancy exclusion, complexity-free). *No valuation word with $\sup_j |S_j| < D^*$ is realizable by a divergent positive orbit — with no hypothesis on factor complexity. This subsumes the Sturmian exclusion ($D < 1$) of the parent program and removes its subexponential-complexity assumption in the divergent case.*

3 The (D, ε) phase diagram

Tilt the strip transfer matrix by the non-coboundary shadow observable $\phi_\omega(b) = \Re(\omega(\zeta_3^{-b} - \mathbb{E}_{\text{geom}}))$ and define

$$\ln \lambda_{\text{eff}}(D, \varepsilon) = \max_{\omega} \inf_{t \ge 0} \left[\ln \lambda_t(D, \omega) - t\varepsilon\right] :$$

the exponential growth rate of D-confined words carrying bias $\ge \varepsilon$. The phase boundary $\lambda_{\text{eff}} = 2$ (Figure 1) runs vertically at D^* for small bias and saturates near $\varepsilon^* \approx 0.54$ for wide bands: *all* narrow bands are excluded, and wide bands tolerate only bias $\lesssim 0.5$. A counterexample's banded windows are therefore confined to the wide-band low-bias region; since the keystone arrow demands a persistent global bias, that bias must be carried by the *band crossings* — the regime of the 2-adic run-length machinery (k consecutive $b = 1$ steps force $x \equiv -1 \bmod 2^{k+1}$), which is the next target.

Remark (Exact certification). *The count $N_D(p)$ is* exactly computable*: D-confinement quantizes B_j to at most $\lfloor 2D \rfloor + 1$ integers in the window $[j \log_2 3 - D,\, j \log_2 3 + D]$, so $N_D(p)$ satisfies a rotation-driven integer recursion over ≤ 3 states per position (for $D < 1.5$), evaluated in exact big-integer arithmetic with no discretization. At $p = 3000$ this gives $\lambda(D) = N_D(p)^{1/p}$ to four digits and refines the threshold to*

$$D^* = 1.2463, \qquad \text{band ratio } 2^{2D^*} = 5.63,$$

and—more importantly—Theorem 1 may be stated with the exact integer $N_D(p)$ in its hypothesis, eliminating Perron estimates from the proof entirely. All other ingredients (repetition rigidity; the Baker cycle bound) are proved in the parent program, the former machine-verified in Lean 4.

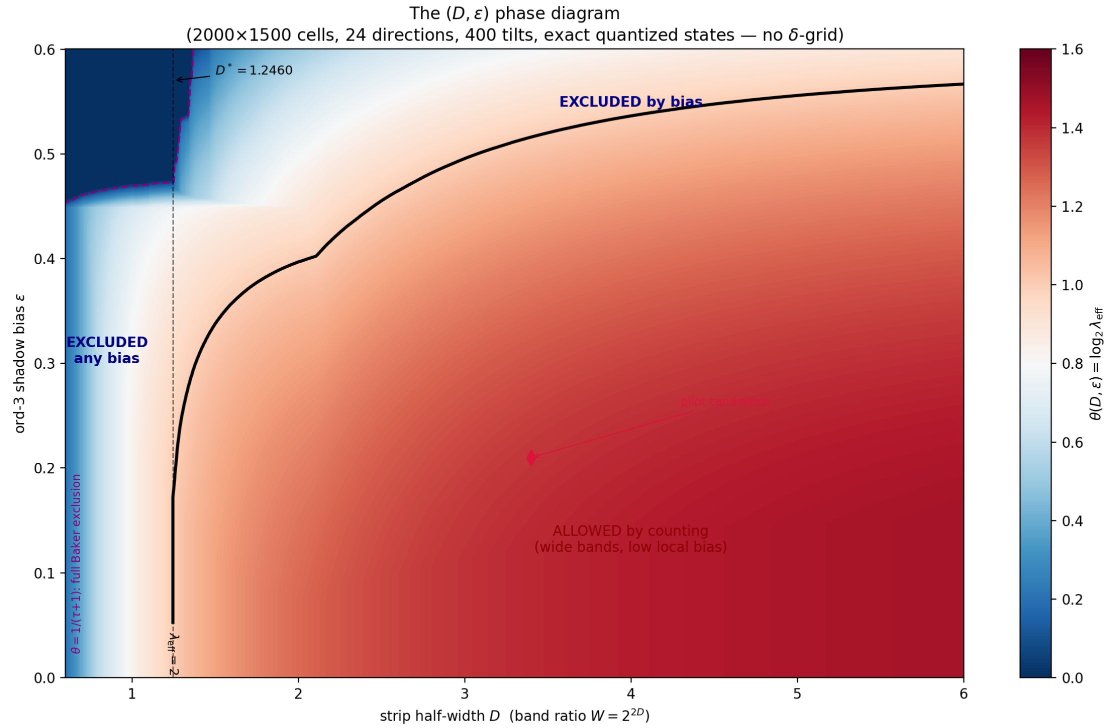

Figure 1: The phase diagram (2000×1500 cells, 24 directions, 400 tilts), computed with exact quantized states — no grid discretization of the state variable. Blue: excluded by counting (repetition forced). Red: allowed by counting; the pilot candidates' windows sit here and die by 2-adic realizability instead.

Remark (Kink structure of the phase boundary)**.** *Computed with exact quantized states, the boundary $\varepsilon^*(D) = \sup\{\varepsilon : \lambda_{\mathrm{eff}}(D, \varepsilon) \geq 2\}$ is smooth except for corners located precisely at the* window-size jumps *$2D \in \mathbb{Z}$, where the number $\lfloor 2D \rfloor + 1$ of integers available to each B_j increments: measured slope jumps in ε^* are -0.089 at $D = 1.5$, -0.015 at $D = 2.0$, -0.020 at $D = 2.5$, and below resolution by $D = 3.5$. The quantization of the confinement window is thus directly visible in the phase boundary, with the new combinatorial freedom absorbed at a decaying rate as the window widens. We caution that δ-grid discretizations of the strip transfer (states = grid points of $[-D, D]$, shifts rounded to the grid) superimpose a spurious resolution-dependent fine structure on the same diagram — scalloped oscillations of typical size $|\Delta\theta| \approx 0.02$, reaching 0.53 at small D — which the exact computation removes entirely; only the corners at $2D \in \mathbb{Z}$ are arithmetic.*

Remark (The binary bias ceiling $\varepsilon_{\mathrm{bin}} = 0.449$)**.** *The diagram contains a second arithmetic line, horizontal and dual to the vertical counting threshold D^*. Confinement pins the Birkhoff mean of any D-confined word at $\log_2 3$, so a word over the binary alphabet $\{1, 2\}$ is forced to the letter frequencies $(2 - \log_2 3, \log_2 3 - 1)$; its shadow average is then a single point on the chord joining ζ_3^{-1} and ζ_3, and the largest bias it can carry is*

$$\varepsilon_{\mathrm{bin}} = \left|(2 - \log_2 3)\, \zeta_3^{-1} + (\log_2 3 - 1)\, \zeta_3 - G\right| = \sqrt{\tfrac{9}{196} + 3\left(\log_2 3 - \tfrac{19}{14}\right)^2} = 0.44903\ldots,$$

with $G = -\frac{2}{7} - \frac{\sqrt{3}}{7} i$ the geometric mean. Since this depends only on calibration, not on D, the line $\varepsilon = \varepsilon_{\mathrm{bin}}$ is exactly horizontal: below it the extremal biased words are binary and $\theta(D, \varepsilon)$ varies smoothly; crossing it forces a positive density of letters $b \geq 3$, each costing drift $b - \log_2 3$ repayable *only by* $b{=}1$ *excursions inside the strip, so for narrow strips the rate collapses almost discontinuously (at* $D = 1.2$*,* θ *falls from* 0.75 *to* 0.15 *across the line; the steepest gradient sits at* $\varepsilon = 0.4489$ *for every* $D \in [1.0, 2.0]$*, fading by* $D \approx 2.5$ *as deep excursions become affordable). Visually the line emanates rightward from* D^* *because to its left it is buried in the region already excluded by counting. Consequence for the program: in any narrow-band window a counterexample's local bias is capped at* ε_{bin} *unless it spends letters* $b \geq 3$ *— precisely the descent-heavy letters taxed by the run-budget clock — so counting capacity exhausts at* D^* *and alphabet capacity exhausts at* ε_{bin}*.*